



INSTRUCTIONS FOR SWITZER FM..RI FLOW SWITCH

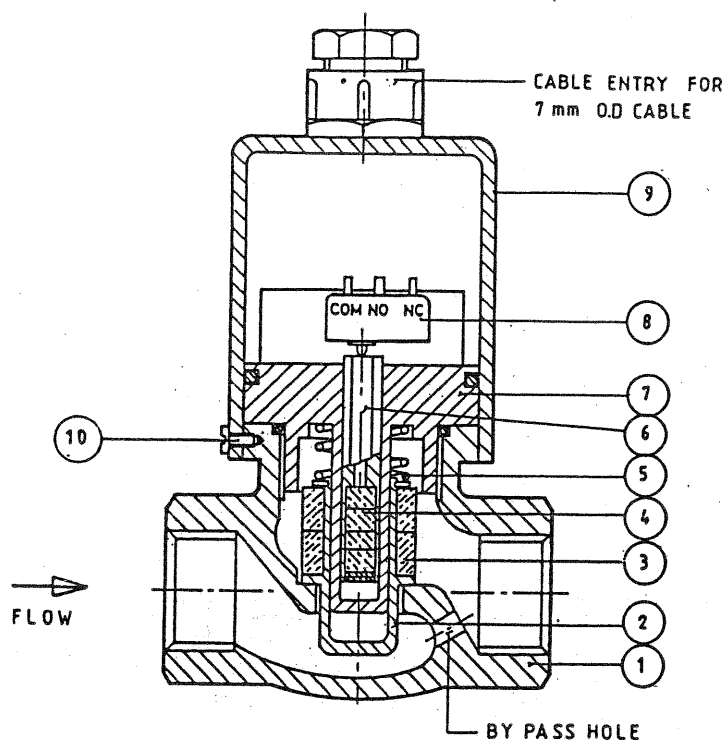
INSTRUCTION FOR INSTALLING OF SERIES FM FLOW SWITCHES

1. Clean the pipe systems prior to the fitting of the Switch to ensure that all scales, dirt, welding residues, pipe trashes are removed.
2. Do not remove the end covers of the Flow Switch until the system is totally ready for installation.
3. Install the Flow Switch ensuring the minimum required straight run to 5D on upstream and downstream sides of Flow Switch respectively.
4. Install the Flow Switch with a suitable Adaptor on line without any mechanical stress on switch body.
5. It is always expedient to mount the Switch in such a way that it can be removed for any maintenance in future.
6. Ensure that the flow direction shown on the body matches that of the flow in the pipe.
7. In case the flow medium has suspended impurities install a strainer in the upstream of the Flow Switch maintaining the straight run requirement of Flow Switch.
8. The Micro Switch is rated for only 5 amp. 220V, max. Hence any attempt to exceed this rating through the contact will permanently damage the Switch. Hence short circuits and over charging of contacts are to be avoided.
9. Secure the fly leads of Micro Switch in terminal box or in any such arrangement to avoid any inadvertent short/open circuiting of lead wires. Adequate care shall be exercised to avoid any mechanical stress on the fly leads.

OPERATION

1. The Switch is generally characterised by fixed set point which can be increased to any admissible value by providing suitable bypass hole. (For more details, consult factory). Please note that under no circumstances, the set point can be reduced below the existing value.
2. Install the Flow Switch with sound installation practices and let in fluid at desired rate of flow. Loosen the doom-nut holding the Micro Switch mounting plastic plate and move the mounting plate up and down to obtain desired change over of contact for increasing / decreasing flow. Once the desired set point is achieved secure the mounting plate with the doom-nut.

Please note that this procedure is only to restore the switch to the factory set value which could have got shifted due to transit / rough handling and the like.



SL.No.	DESCRIPTION	QTY.
1	BODY - IN LINE	1
2	PISTON ASSY.	1
3	MAGNETS - PISTON	2
4	MAGNETS - PLUNGER	4
5	COMP. SPRING	1
6	PLUNGER ASSY.	1
7	PISTON HOUSING	1
8	MICRO SWITCH	1
9	SWITCH HSG COVER	1
10	COVER FIXING SCREW	1

MAINTENANCE

The microswitch housing is totally separated from the process media and actuating piston mechanisms with a magnetic coupling. No maintenance is normally required except in cases where the process fluid might have foreign particles with iron content. If the particles get stuck to the ring magnets attached to the moving piston there are chances of the piston not coming back to its normal position. In that event, even if the flow stops the microswitch contacts would not change over due to piston not coming back to rest on its seat.

Periodical cleaning of the piston magnet assembly would then be required. The periodicity has to be determined depending upon the fluid conditions.

Cleaning procedure

Remove the cable gland. Remove the cover fixing screw(10). Pull out the top housing carefully holding the cable without any force acting on it. Using a 35mm 'C' spanner, unscrew the piston housing(7) completely. Pull apart the piston magnet assembly(2) with light force, clean it thoroughly and put back. Clean the inner surfaces of the body(1) including the bypass hole (if any).

Now the piston housing(7) can be threaded back into position, switch housing cover(9) can be refixed and cover fixing screw(10) back in position.

Please ensure that fixing screws of microswitch(8) are not disturbed and it is in proper contact with the plunger(6).

Switzer

Works:

127 Sidco Estate, Ambattur
Madras 600 098
Ph: 6242244, 6242255
Fax: 6258739
E-mail: switzer@eth.net

Sales:

9 South Boag Road, PB 1423
T Nagar, Madras 600 017
Ph: 4340999, 4343956
Fax: 4347887
E-mail: switzer@switzerindia.com